

The shape of a cubic



1

- a They become very large in the positive direction
- b They become very large in the negative direction

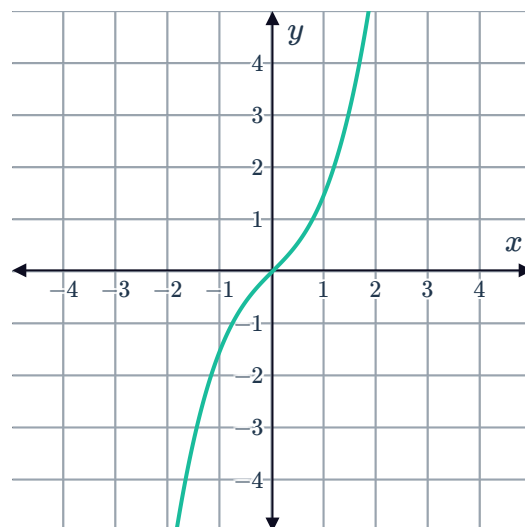
2

- a Negative
- b $(0, 3)$
- c $y = -2x^3 + 3$

3

- a Positive

b



- c $(0, 0)$

4 $(0, 0)$

5

- a C
- b A
- c B

6

- a
 - i $x > 0$
 - ii $x < 0$
 - iii $(0, 3)$
- b
 - i $x > 3$
 - ii $x < 3$
 - iii $(3, 2)$

7

- a $x = -2$
- b $y = 4$

8

- a Negative
- b $(0, 3)$
- c $x < 0$
- d $x > 0$

9 $x \geq 3, x = -1$



Transformations

10

a No

b No

c No

d Yes

11

a $(0,0)$

b $(0,3)$

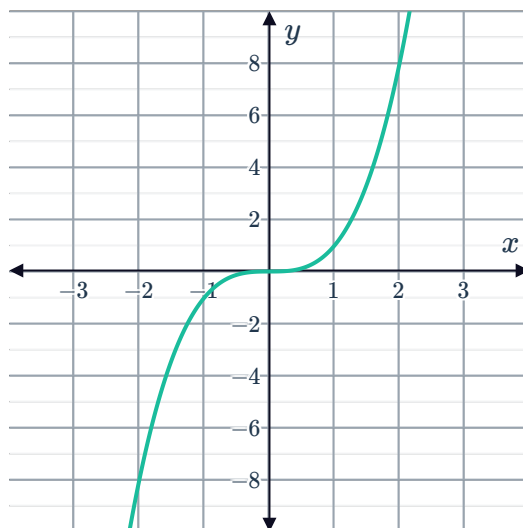
c $(0,4)$

12

a
i

x	-2	-1	0	1	2
y	-8	-1	0	1	8

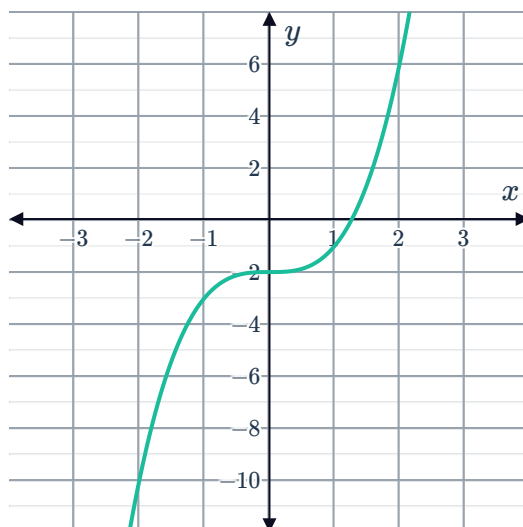
ii



b
i

x	-2	-1	0	1	2
y	-10	-3	-2	-1	6

ii



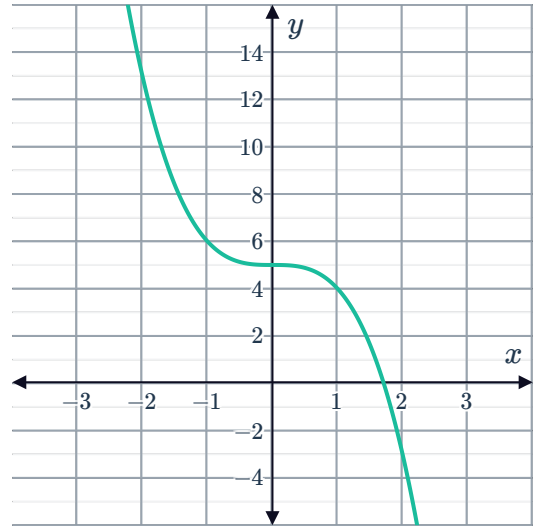
c

i

x	-2	-1	0	1	2
y	13	6	5	4	-3



ii



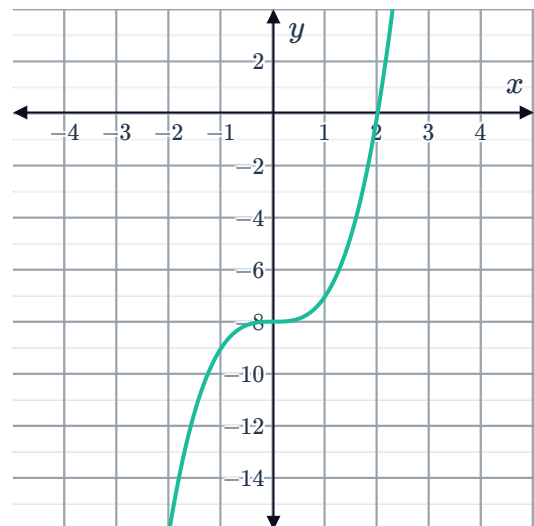
13

a $x = 2$

c $(0, -8)$

b $y = -8$

d



14

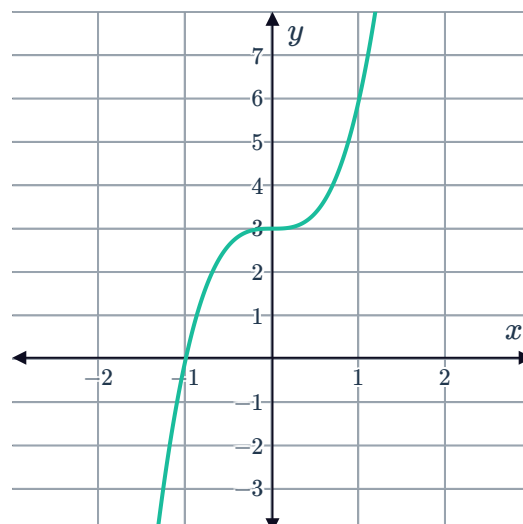
a $x = -1$

c $(0, 3)$



b $y = 3$

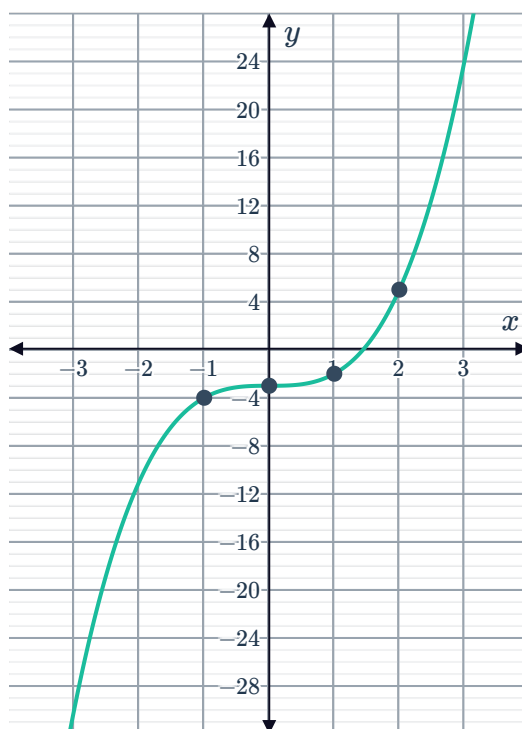
d



15

a $A(-3, -30), B(-2, -11), C(-1, -4), D(0, -3), E(1, -2), F(2, 5), G(3, 24)$

b



16

a $y = -4$



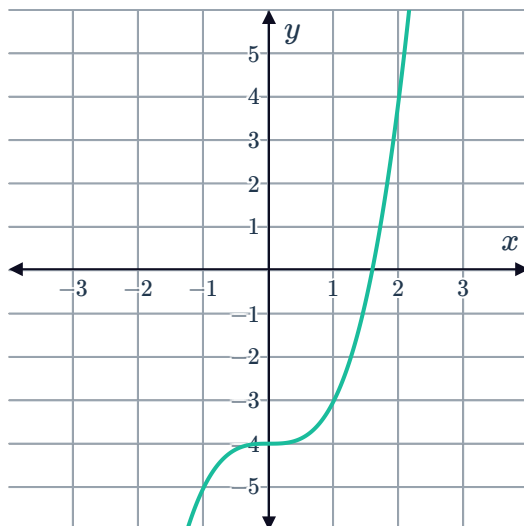
b

x	-1	1	2
y	-5	-3	4

c $(-\infty, \infty)$

d $(-\infty, \infty)$

e

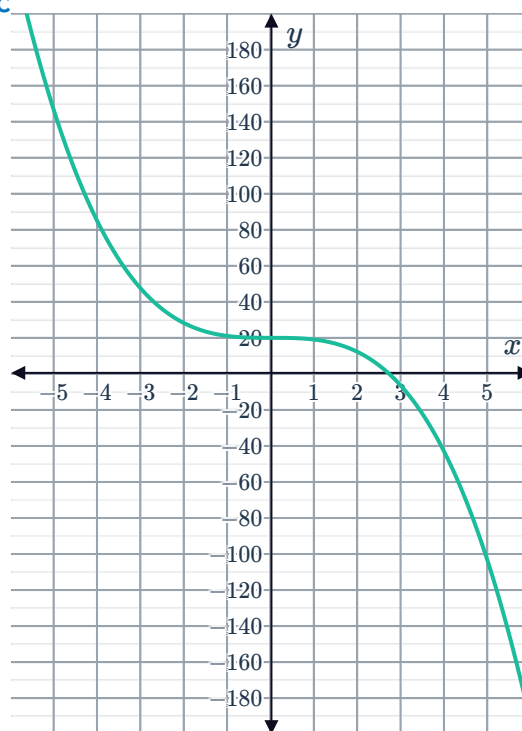


17

a $(2, -4)$

b $(6, -196)$

c



18

a $x = 2$

e $(-\infty, \infty)$

b $y = 16$



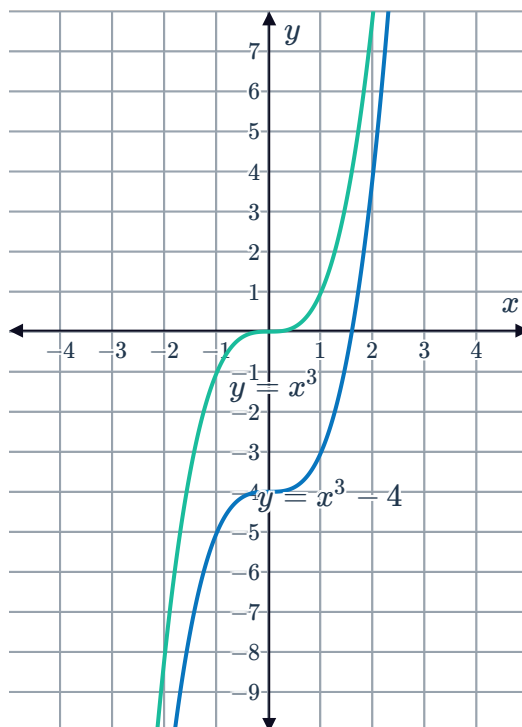
c $(0, 16)$

d $(-\infty, \infty)$

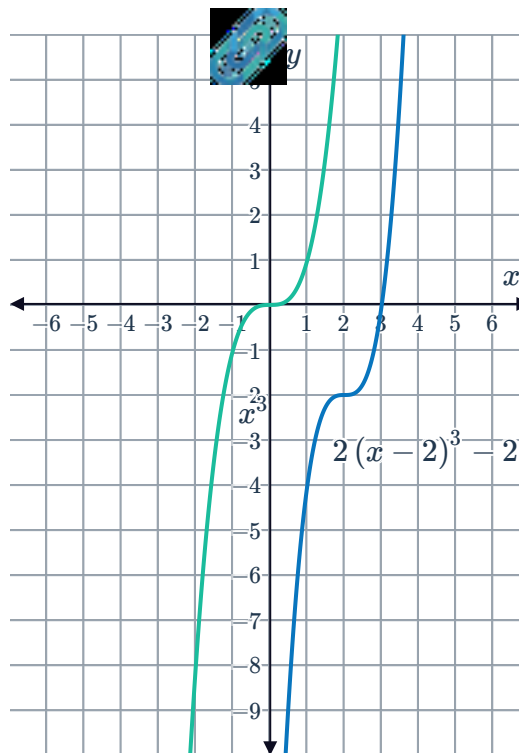
f



19



20



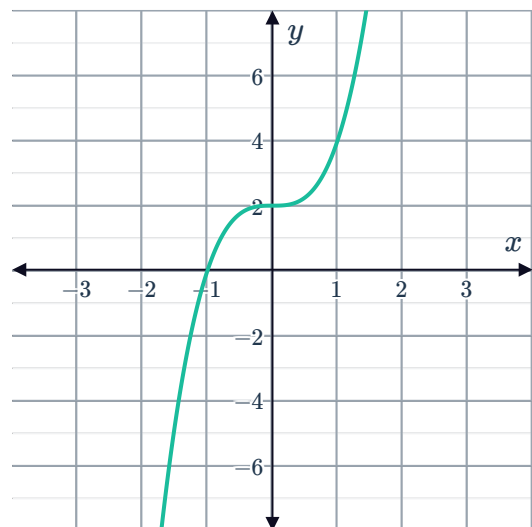
21

a

- i Increasing
- iii (0, 2)

ii More steep

iv



b

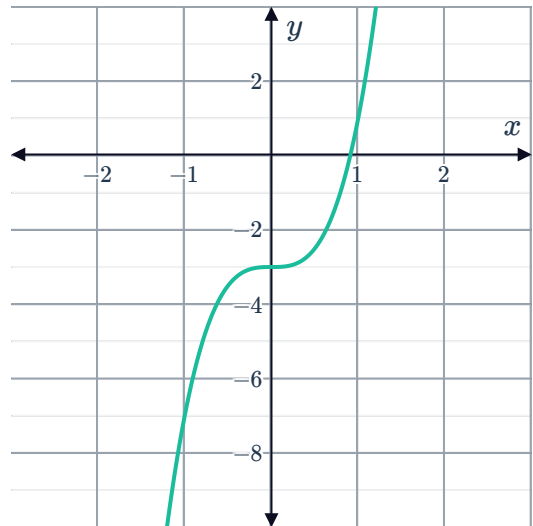
i Increasing

iii $(0, -3)$



ii More steep

iv



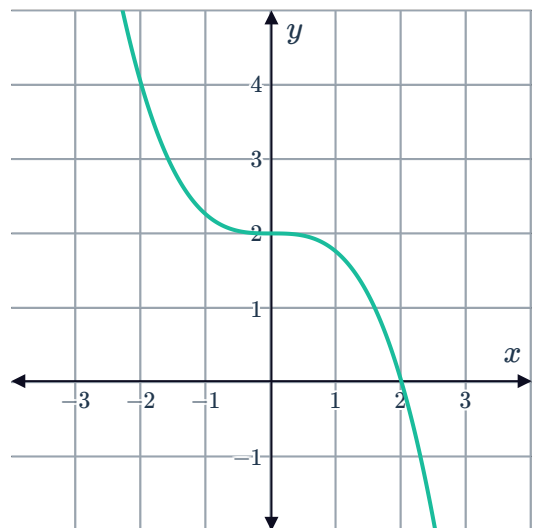
c

i Decreasing

iii $(0, 2)$

ii Less steep

iv

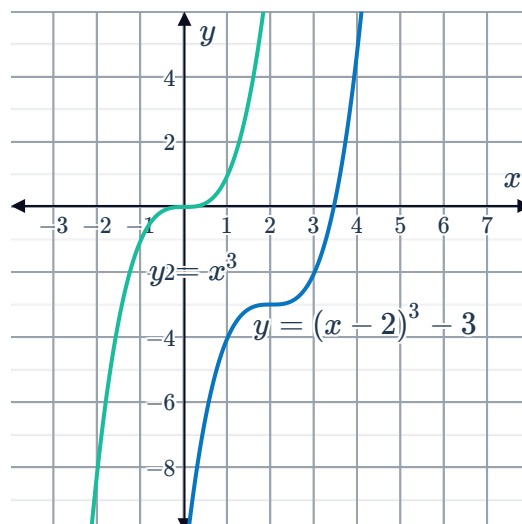


22



a Move the graph to the right by 2 units and down by 3 units.

b



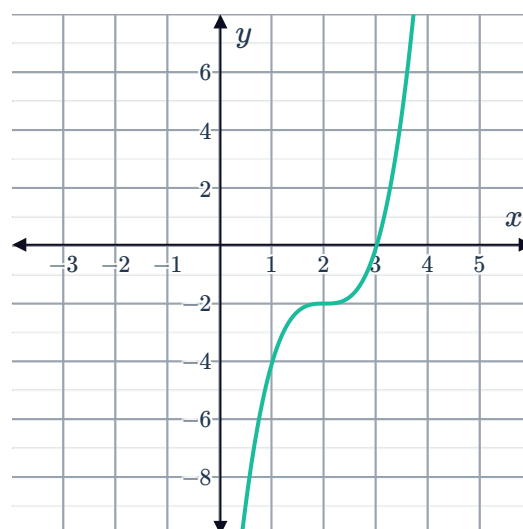
23

a Increasing

c $(2, -2)$

b More steep

d



24

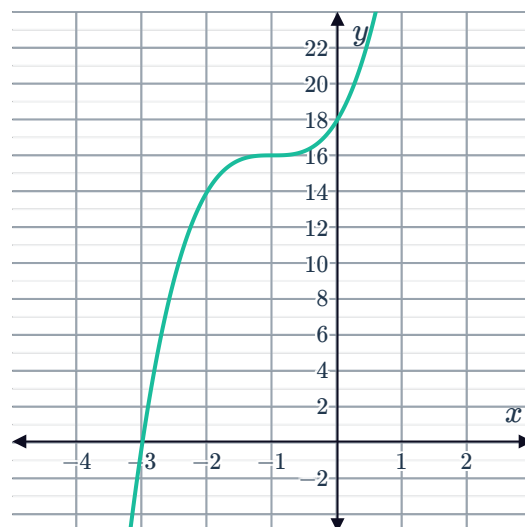
a $x = -3$

c $(-1, 16)$



b $y = 18$

d



25

a $x = 2$

c $(1, 3)$

b $y = 6$

d



Factored form of a cubic function

26 3

27 3

28

a No

b No

c No

d Yes

29

a No

b Yes



c No

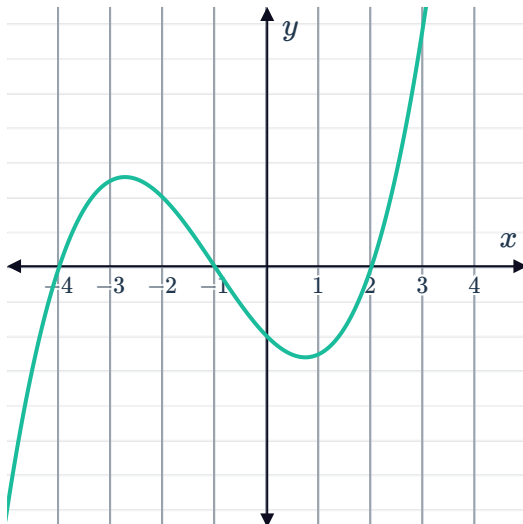
d No

30

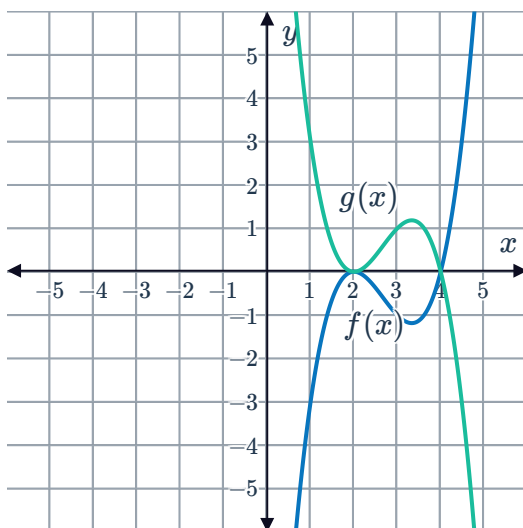
a $x = -3, 2, 5$

b $y = -(x + 3)(x - 2)(x - 5)$

31



32



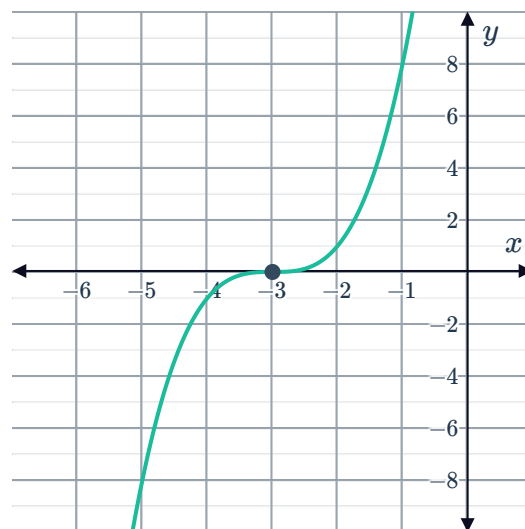
33

a

x	-5	-4	-3	-2	-1
y	-8	-1	0	1	8



b

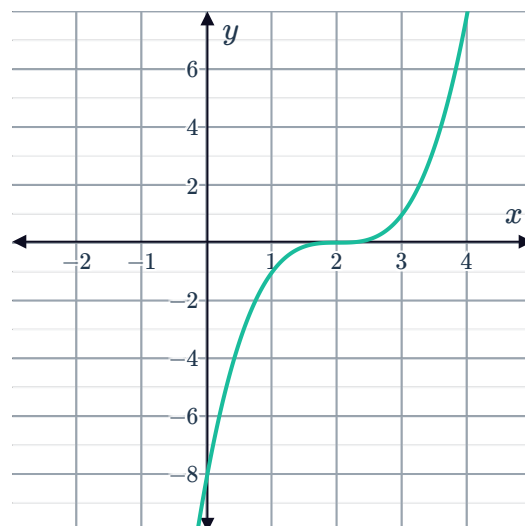
c $(-\infty, \infty)$ d $(-\infty, \infty)$

34

a

x	0	1	2	3	4
y	-8	-1	0	1	8

b



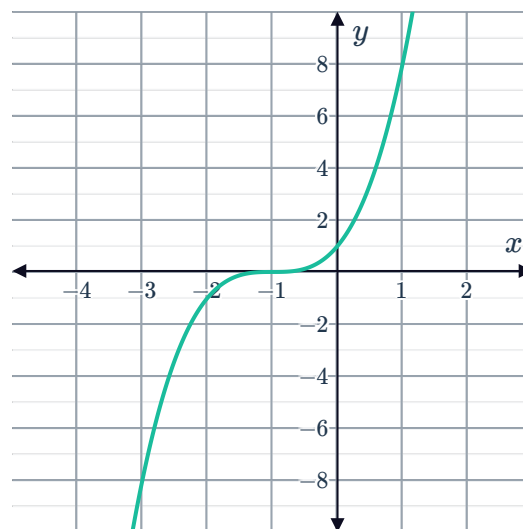
35

a

x	-3	-2	-1	0	1
y	-8	-1	0	1	8



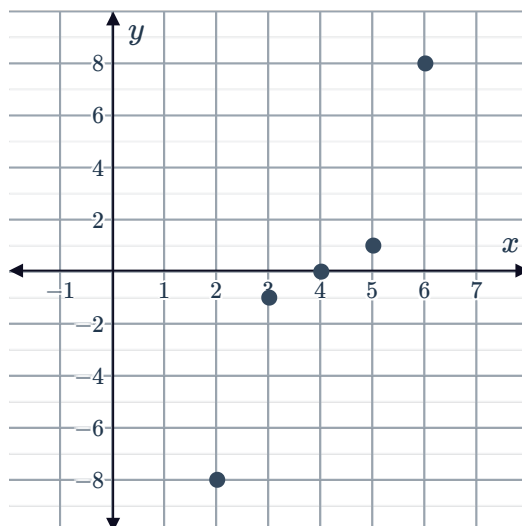
b



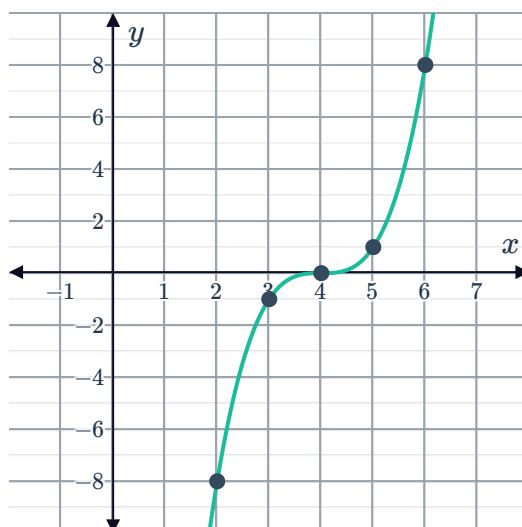
36

a $A(3, -1), B(2, -8), C(5, 1), D(4, 0), E(6, 8)$

b



c

d $8 < y < 27$

37

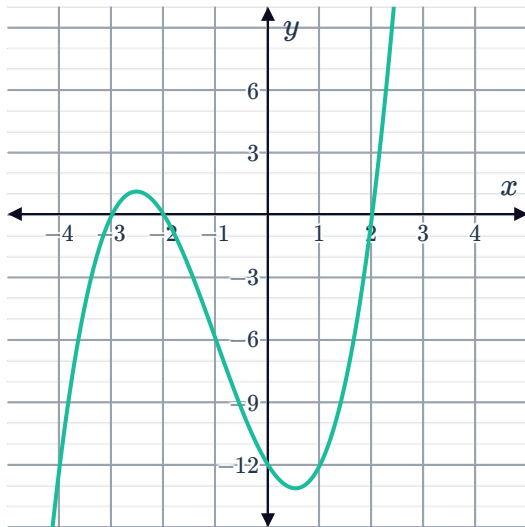


a

i $x = -3, -2, 2$

ii $y = -12$

iii

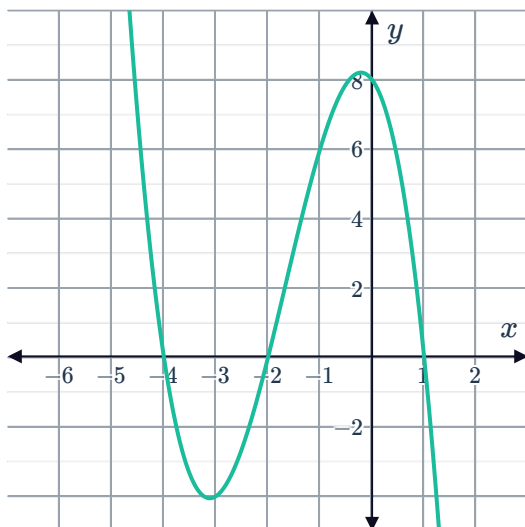


b

i $x = -4, -2, 1$

ii $y = 8$

iii



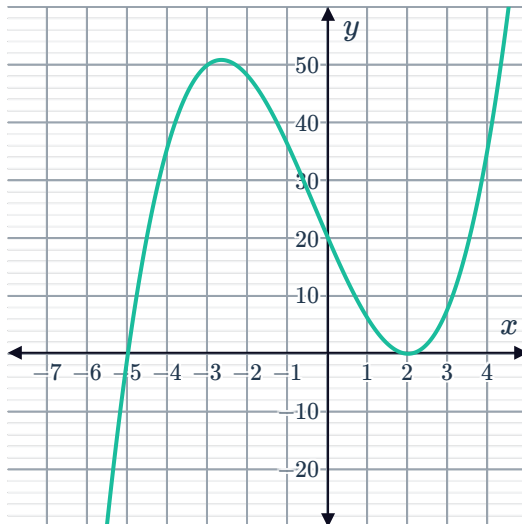
c

i $x = 2, x = -5$



ii $y = 20$

iii



38

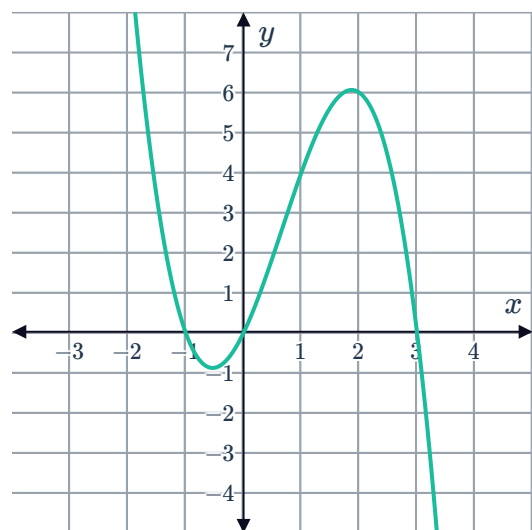
a

i $-x(x - 3)(x + 1)$

iii $x = 0, 3, -1$

ii $y = 0$

iv



b

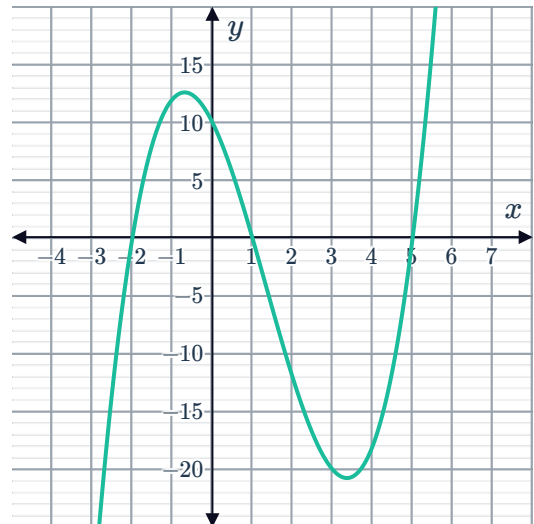
i $(x + 2)(x - 5)(x - 1)$

iii $x = -2, 5, 1$



ii $y = 10$

iv



Applications

39

a

r	1	2	4	5	7
V	$\frac{4\pi}{3}$	$\frac{32\pi}{3}$	$\frac{128\pi}{3}$	$\frac{500\pi}{3}$	$\frac{1372\pi}{3}$

b

i No

ii Yes

iii No

iv No

c $r = 6$ m

40

a $k = 5$

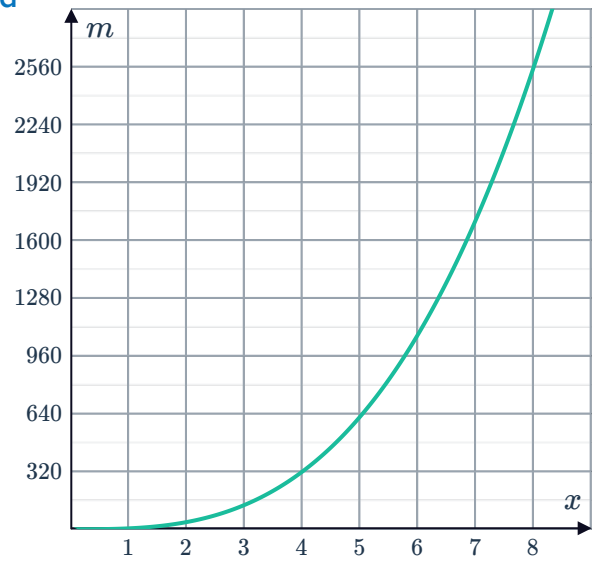
c

x	1	2	3	4
m	5	40	135	320



b $m = 5x^3$

d



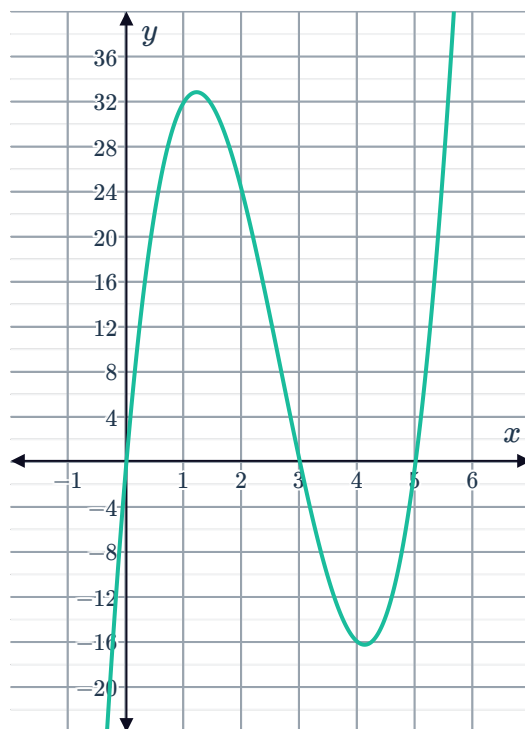
e 3071 g

f 7 cm

41

a $V = 4x(3-x)(5-x)$

c



b $0 < x < 3$

d 24 cm^3

42

a $x^2 + h^2 = 15$

c $0 < h < \sqrt{15}$

b $V = 2\pi h(15 - h^2)$

d $V = 20\sqrt{5}\pi$